Muhammad Muizz Zafar

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Software Engineer | ETL Pipelines | Data Platforms | Cloud Infrastructure

Skills

Programming Languages:	Java, Python, TypeScript, GoLang, SQL, JavaScript, Kotlin, C#, C++, HTML/CSS, PHP
Tools/Frameworks:	AWS, AWS CDK, CloudFormation, Docker, Splunk, CloudWatch, Athena, Apache Spark, Amazon

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Software Developer — Jan 2025 - Present
Kinaxis Inc. — Ottawa, ON

- Developed an automation framework to convert legacy KUnit tests written as sql queries to **JUnit** in **Java** by adding serilization modules at the start and end of each backend algorithm to capture the inputs and outputs for each test.
- Implemented **Java services** supporting scope control to detect incompatiblities between different features of the Rapid Response product which was used by customers and reduced L3 escalations by customers.
- Designed automated queries in SQL to detect known behavioral differences across different algorithms between the legacy and modular codebase.
- Used **Javascript** to write automated workflows within the rapid response product to run the known difference queries and visualize the known differences for higher management, directors, and customers to see the differences between legacy and modular codebase

ETL Developer — May 2024 - December 2024
Elections Canada — Ottawa, ON

- Built automated **Python ETL validation pipelines** and **Azure DevOps CI/CD workflows** to verify high-volume production data movement across multiple environments and multi-table workflows, reducing manual validation effort.
- Used **Apache Spark** to profile, transform, and validate large datasets during pipeline verification, accelerating bulk data checks and anomaly detection across staging and production environments.
- Built internal **data analytics dashboards** using **Python, Streamlit, and Plotly Dash** to surface pipeline health, operational metrics, validation trends, and data quality issues for production systems.
- Coordinated version-controlled database deployments and release processes, improving traceability and reliability of **production data infrastructure** across shared environments.
- Collaborated in an **Agile cross-functional team** with cloud engineers and DevOps teams to improve **distributed data workflows**, release quality, and production reliability.

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze uploaded client files and images and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Defined and maintained cloud infrastructure using **TypeScript**, **AWS CDK**, and **CloudFormation** to provision monitoring resources and improve repeatability of service deployments.
- Implemented concurrent processing logic in **GoLang** using goroutines to improve throughput, reliability, and fault handling for **distributed cloud data processing** workloads.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring across large volumes of cloud artifacts.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates with less manual operational effort.
- Supported large-scale data processing workflows with **Amazon EMR** and **Apache Spark**, using batch jobs to analyze cloud metadata and improve downstream reporting, validation, and metrics generation.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track system logs, metrics, and failures, improving observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to validate cloud services and improve reliability of distributed serverless data pipelines.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Built **Docker** containers and automated event ingestion into **PostgreSQL** using **Python** and **GraphQl**, reducing testing time and improving deployment readiness.